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NSE

PRICE PREDICTION

BUY

SELL

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Problem Statement

Retail investors on the NSE often face information asymmetry, high volatility, and liquidity challenges when deciding at what price to place buy (bid) or sell (offer) orders. Many rely on **gut feel**, **broker advice**, or **simple charts**, leading to suboptimal entries/exits, slippage, missed opportunities, or unnecessary losses.

The system aims to reduce this gap by using historical daily data (2022–2025) to forecast near-term price behavior (e.g., next-day close, Day Low, or Day High). This helps investors set smarter limit orders, for example, bidding near a predicted Day Low or offering near a predicted Day High, while incorporating volume and market context.

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Data Understanding

The data is sourced from the Mendeley Data repository(open research data repository) which consists 63,059 daily trading rows compiled across all listed stocks on the NSE from **January 3, 2022, to October 31, 2025**

Input Variables

1. return_lag1, return_lag2- last few days' price changes
2. return_roll5_mean, return_ewm5 -average patterns over recent days
3. price_pos_52w - how high/low the current price is relative to its yearly range
4. volume_ratio - current trading activity compared to recent average
5. dow_num -day-of-week patterns in trading behavior

Target Variable

1.next_day_price - model predicts tomorrow's closing price

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Stakeholders

Retail Investors (Primary Beneficiary)

These are the everyday Kenyan individuals who buy and sell stocks on the NSE through brokers or mobile platforms like Safaricom's Ziidi Trader. They are the direct end-users of the prediction system.

Stockbrokers & Investment Advisors

Licensed NSE brokers act as intermediaries between investors and the exchange. They advise clients on entry and exit prices daily. This system is directly relevant to them as a value-added advisory tool

NSE

The NSE as an institution has a vested interest in projects that increase retail participation and market confidence. Higher participation means greater trading volumes, more listing activity, and stronger exchange revenues.

Capital Markets Authority (CMA)

The CMA is Kenya's financial markets regulator responsible for investor protection and market integrity. Any system that helps retail investors make more informed decisions aligns directly with the CMA's mandate.

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Exploratory Data Analysis

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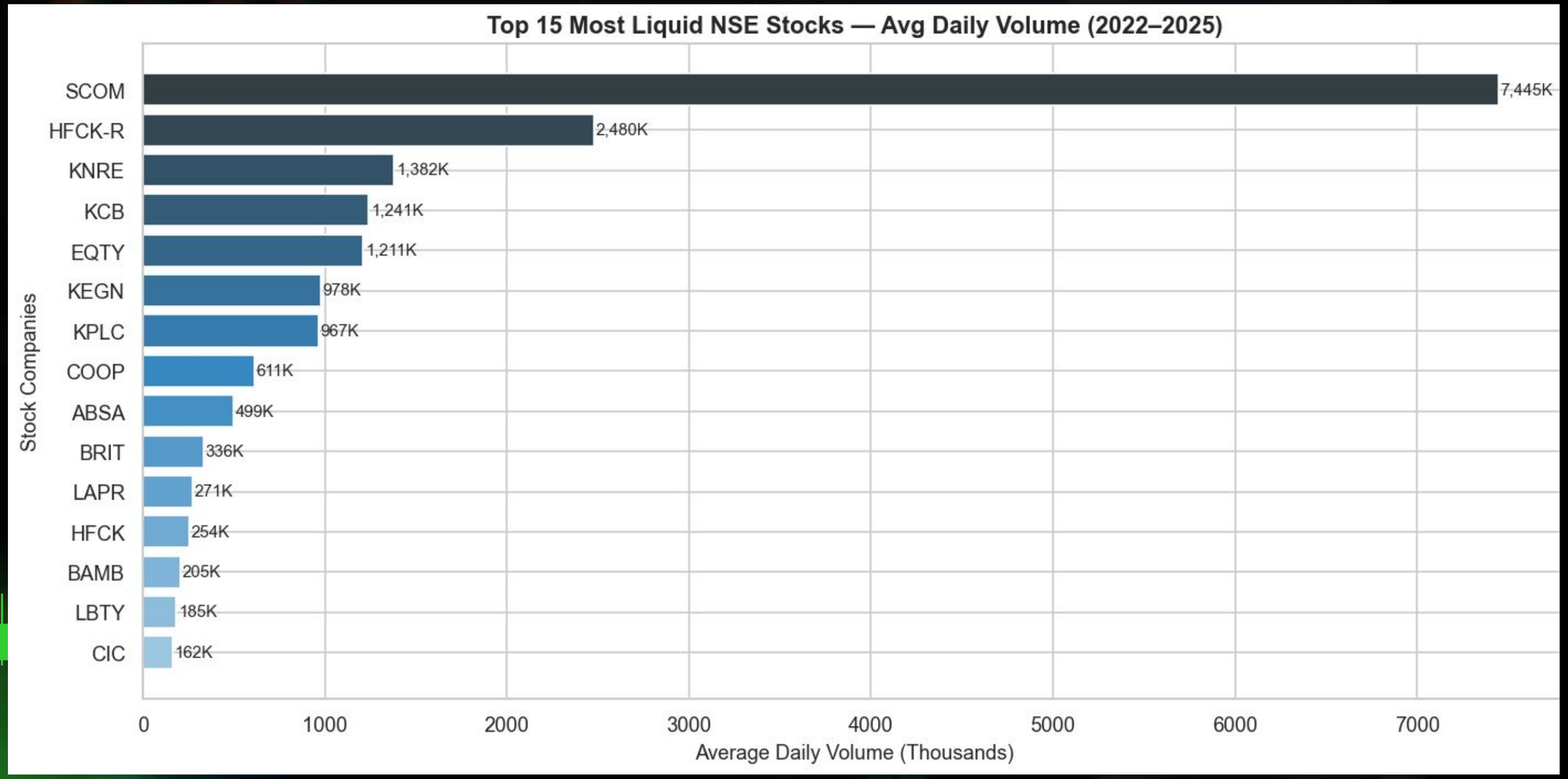
Most Liquid NSE Stocks



Studio Shodwe

Findings

1

SCOM (Safaricom) dominates entirely at 7,445,002 shares per day, more than 3 times the volume of the second-placed stock HFCK-R. Safaricom is Kenya's most traded stock by far.

2

KCB (1,241,449) and EQTY (1,211,458) are tied as the two most liquid banks, both ideal for modelling because they have consistent, high-volume trading every day.

3

KEGN (978,047) and KPLC (966,687) representing KenGen and Kenya Power are surprisingly liquid, reflecting retail investor interest in utility stocks.

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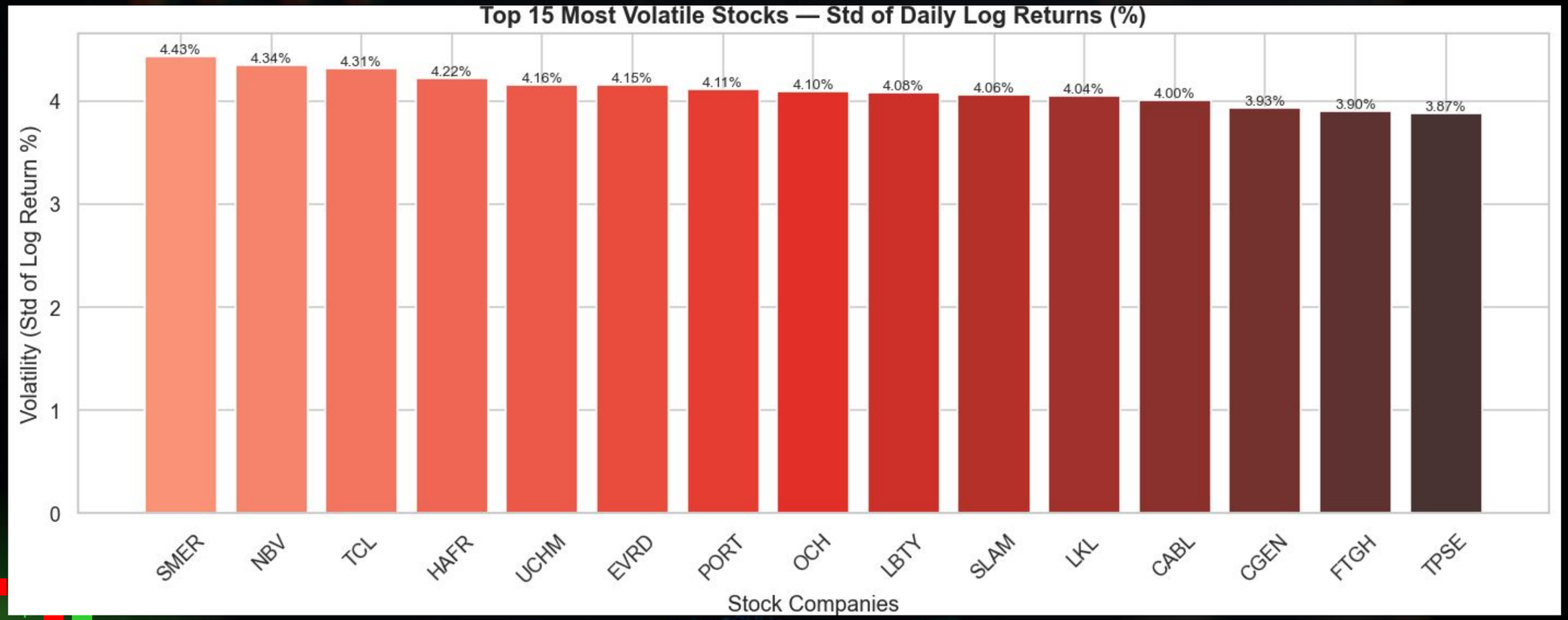
Most Volatile NSE Stocks



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Findings

1

SMER leads at 4.43% daily volatility, meaning on any given day, this stock's price can swing by more than 4% in either direction on average. For a retail investor, this is extremely risky without a prediction system.

2

All 15 stocks here are small-cap or mid-cap, none of the blue chips (EQTY, KCB, SCOM, SCBK) appear. This is expected. Large, heavily traded stocks have tighter bid-ask spreads and more price stability. Small illiquid stocks swing wildly because a single large trade can move the price significantly.

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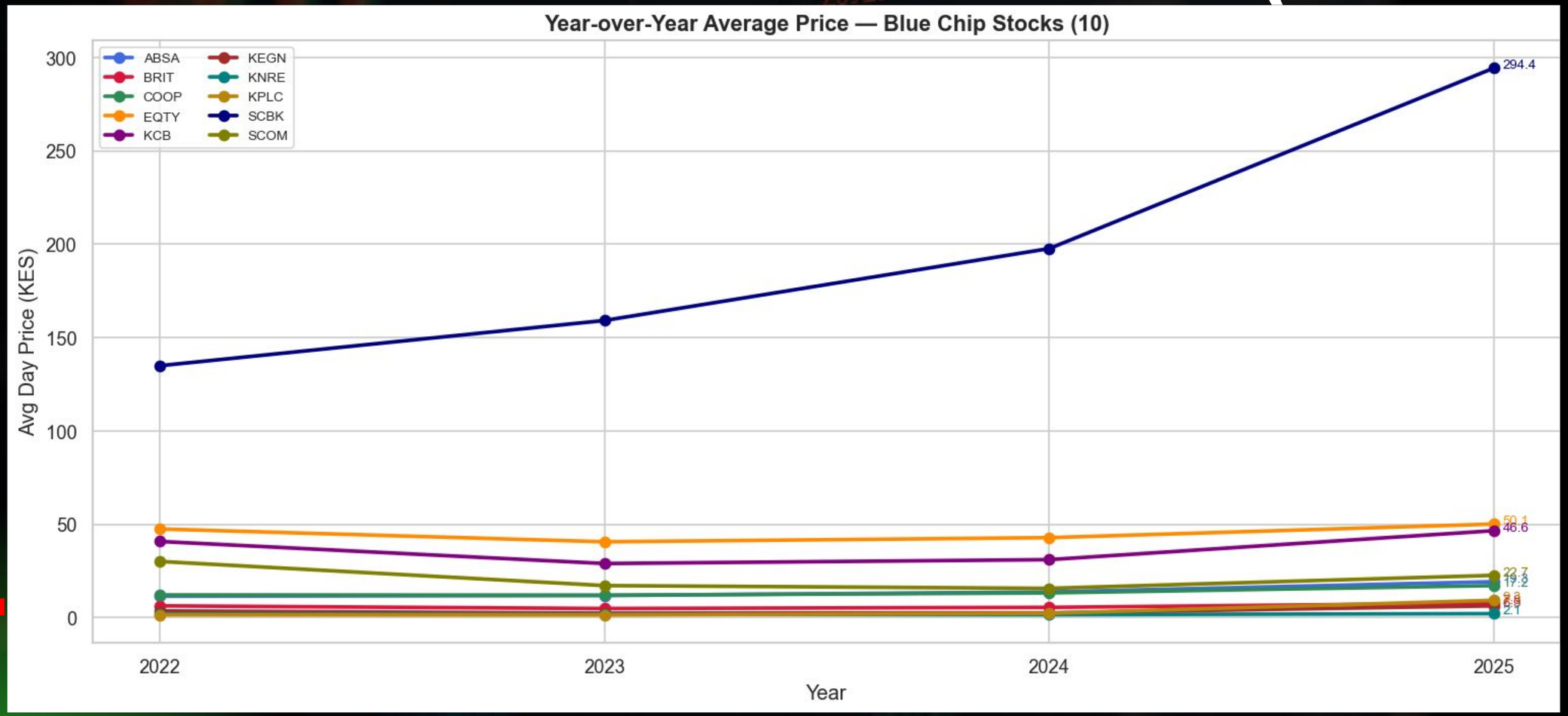
Year-Over-Year Price Trends (Blue



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Findings

1

SCBK (Stanbic Bank) is the standout performer of the entire dataset. Its price nearly doubled from KES 134.93 in 2022 to KES 294.41 in 2025, a gain of 118%. This is driven by strong earnings growth and foreign investor interest in Kenyan banking stocks.

2

SCOM (Safaricom) fell from KES 30.12 in 2022 to just KES 15.70 in 2024, a 48% decline driven by Ethiopia launch losses and rising interest rates reducing growth stock valuations. It has partially recovered to KES 22.68 in 2025.

3

ABSA and COOP both show consistent, steady upward trends throughout all four years, suggesting more defensive, predictable price behaviour exactly the kind of stocks where prediction models perform best.

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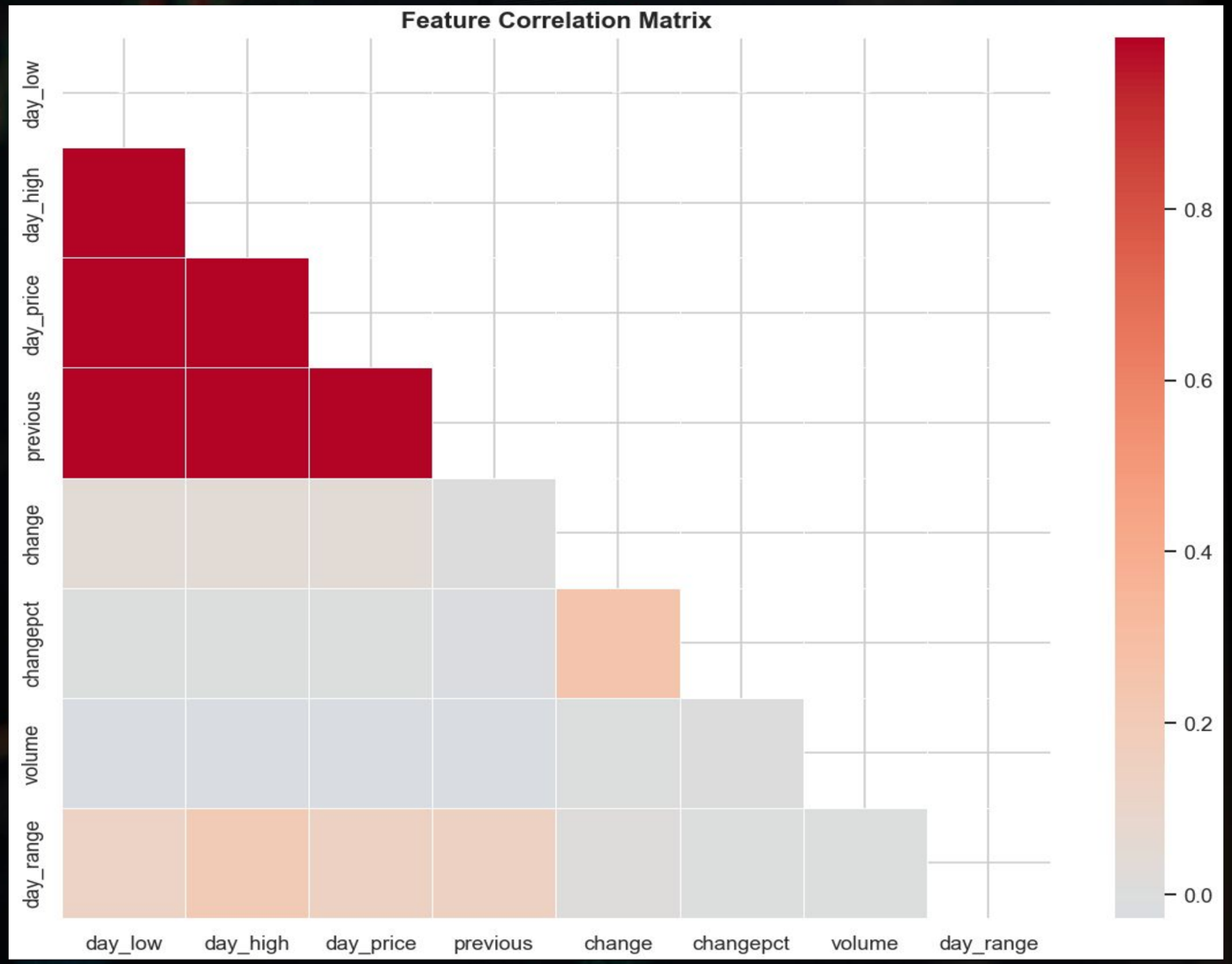
Correlation Matrix



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Findings

day_low (1.000) and day_high (0.998) have near-perfect correlation with day_price which makes complete sense since all three move together. A stock priced at KES 50 will have a Low around KES 49 and a High around KES 51.

previous (0.999) is the breakthrough finding. Yesterday's closing price explains 99.9% of today's price level. This is a strong signal where a stock closed yesterday is by far the strongest signal for where it will trade today.

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Statistical Modeling

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Baseline 1: Persistance

- Predicts tomorrow's price = today's price
- Extremely strong on NSE blue chips (avg daily move < 2%)
- Reveals: Prices behave close to a random walk at daily frequency

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Baseline 2: Rolling 5 Day Mean

- Uses average of last 5 trading days
- Performed worse than persistence. Features need to be more informative than lagged prices
- Reveals: No strong short term momentum

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Baseline 3: Exponentially Weighted Mean (EWM)

- Similar to rolling mean but gives more weight to recent prices. Determines whether yesterdays price matters more than last weeks price

- Still failed to beat persistence

- Reveals: Recent price history alone carries very weak predictive signal. The market does not have strong short term memory

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Key Takeaways

- Market moves very little day to day
- There is no consistent short term trend
- Recent price history carries weak signal
- Absolute price levels are unstable across time hence the reason for predicting returns instead

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Modeling Development

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Model 1: Ridge Regression

- We implemented Linear regression with L2 regularisation (alpha=100) to handle the high correlation between lag features.
- Training was done on all 17 engineered features.

- **Validation METRICS:**
  MAE: 0.529 KES | MAPE: 2.62%
- The dominant coefficients were low_lag1 (0.44), high_lag1 (0.43), and price_lag1 (0.31) - the model is essentially estimating tomorrow's price as a weighted average of yesterday's trading range.
- We found that Model 1 does not meaningfully improve on persistence.

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Model 2: Model 2 - ARIMA (per stock)

- We ran the Augmented Dickey-Fuller test on each stock's price series to check for stationarity.
- Three stocks (ABSA, COOP, KPLC) were stationary at the 5% level; the remaining seven required first differencing.
- We then selected the best ARIMA order per stock using AIC grid search over $p \in \{0,1,2\}$ and $q \in \{0,1,2\}$.

Stock	Best Order	AIC
ABSA	ARIMA(1,0,2)	−388.9
BRIT	ARIMA(0,1,1)	−615.5
EQTY	ARIMA(1,1,0)	969.0
KCB	ARIMA(0,1,1)	767.3
KEGN	ARIMA(2,1,2)	−1850.2

Average validation MAE across blue chips:
4.77 KES | MAPE: 21.7% |RMSE : 8.3

ARIMA performs poorly in this setting.
It is univariate i.e, it ignores volume, the 52-week position, and all the other features we built.
It also has no mechanism to handle the structural price jumps seen in KPLC and KEGN, leading to large sustained errors.

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Model 3 - XGBoost (Walk-Forward)

Walk-forward XGBoost trains on all available history up to day t, predicts day t+1, then expands the window by one and repeats. This is the most realistic evaluation setup for a sequential forecasting problem.

Results per stock (validation):

Stock	MAE	vs Persistence	Dir. Accuracy
ABSA	0.135	−0.008	38.4%
EQTY	0.411	−0.022	35.9%
KCB	0.416	−0.027	39.4%
KEGN	0.053	competitive	44.2%

Directional accuracy (this is: whether we correctly predict up vs down) ranges from 36–50%; barely better than a coin flip. This is expected: predicting the direction of a daily return is much harder than predicting the price level.

We also extended XGBoost to predict a price range (next-day low and high) per stock. Coverage — the proportion of days where the actual price fell inside the predicted range:

A coverage above 80% means the model's predicted band contains the actual price on more than 4 out of 5 trading days, which is useful for limit order placement.

Stock	Coverage
ABSA	83.2%
COOP	81.7%
EQTY	78.1%

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Model 4 - LightGBM

LightGBM trained on all 50 engineered features, predicting next-day day_price across all blue chips pooled together.

Set	MAE	MAPE
Validation	3.45 KES	—
Test	13.16 KES	7.13%

The large gap between validation and test MAE signals that LightGBM overfit to 2024 patterns and did not generalise well to 2025.

The structural price changes in KPLC and KEGN in 2025 are likely responsible — no model trained on 2022–2023 data could anticipate a 4× price jump.

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Model Comparison

Model	Validation MAE (KES)	Notes
Persistence	0.511	Hardest to beat
Rolling 5-day	0.712	Worse than persistence
EWM span=5	0.673	Worse than persistence
Ridge Regression	0.529	Marginal, not significant
ARIMA	4.767	Poor, ignores features
XGBoost Walk-Forward	~0.14–0.42	Competitive per stock
LightGBM	3.449	Overfit to 2024

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Recommendations

1.Include macroeconomic data (CBK exchange rates,inflation rates) to improve model performance.

2.Explore deep learning models (e.g., LSTM) to capture more complex patterns.

3.Add more features such as sentiment or broader market indicators such public mood states,social media volume

4.Incorporate event-based data (e.g., earnings releases, dividends, policy announcements) to capture sudden market movements.

5.Use ensemble methods to combine strengths of different models

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Limitations

1.The model relies mainly on historical price and volume data and does not account for macroeconomic factors such as CBK exchange rates,inflation rates and external shocks such as news, policy changes, or corporate events.

2.Strong correlation between variables such as day price, low, and high reduces feature diversity.

3.The model struggles to significantly outperform simple baseline approaches

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Limitations

4.High market volatility in stocks introduces noise, making short-term predictions less stable.

5.Some stocks have low or zero trading activity, affecting data reliability.

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Deployment

https://nseblue-chipsstockspredictor.streamlit.app

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Q&A

Session

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Ready to Invest Now?